

GIS-BASED AGRICULTURAL LAND SUITABILITY ANALYSIS OF KAVREPALANCHOK DISTRICT

A MINI-PROJECT REPORT

SUBMITTED FOR

INTERNAL EVALUATION OF

GEOGRAPHIC INFORMATION SYSTEM

(GEOM 204)

BE IN GEOMATICS ENGINEERING, SEMESTER IV



KATHMANDU UNIVERSITY

SCHOOL OF ENGINEERING

DEPARTMENT OF GEOMATICS ENGINEERING

Submitted By:

Samir Basnet (032348-22)

Sugam Krishna Dhungana (032357-22)

Submitted To:

Er. Anish Bhandari

Lecturer, DOGE

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1. INTRODUCTION

Agriculture is the backbone of Nepal's economy, providing livelihoods for a significant portion of the population. In Kavrepalanchok District, agriculture is important in sustaining rural communities and ensuring food security. However, the region faces numerous challenges, including unpredictable weather patterns, and limited availability of suitable agricultural land. These challenges require a systematic approach to identify and optimize the use of land resources for sustainable farming practices. Agricultural land suitability analysis is a critical tool for identifying the most appropriate areas for specific crops based on factors such as soil quality, topography, climate, and water availability. By assessing land suitability, farmers and policymakers can make informed decisions to optimize land use, enhance productivity, and promote sustainable farming practices. Geographic Information Systems (GIS) have emerged as a powerful tool for conducting such analyses, enabling the integration and visualization of spatial data to support decision-making.

2. STUDY AREA

Kavrepalanchok District is situated approximately 30-40 kilometers east of Kathmandu, the capital city of Nepal. It is bordered by Sindhupalchok District to the north, Ramechhap District to the east, Lalitpur District to the west, and Dolakha District to the northeast. The district covers an area of 1,396 square kilometers and is divided into 13 local administrative units. The district is located between latitude and longitude of 27.45° N - 27.75° N and 85.45° E - 85.85° E respectively. The district's elevation ranges from approximately 280 meters in the lower river valleys to around 3,018 meters at Bethanchowk Narayan Danda, the highest peak in the district.

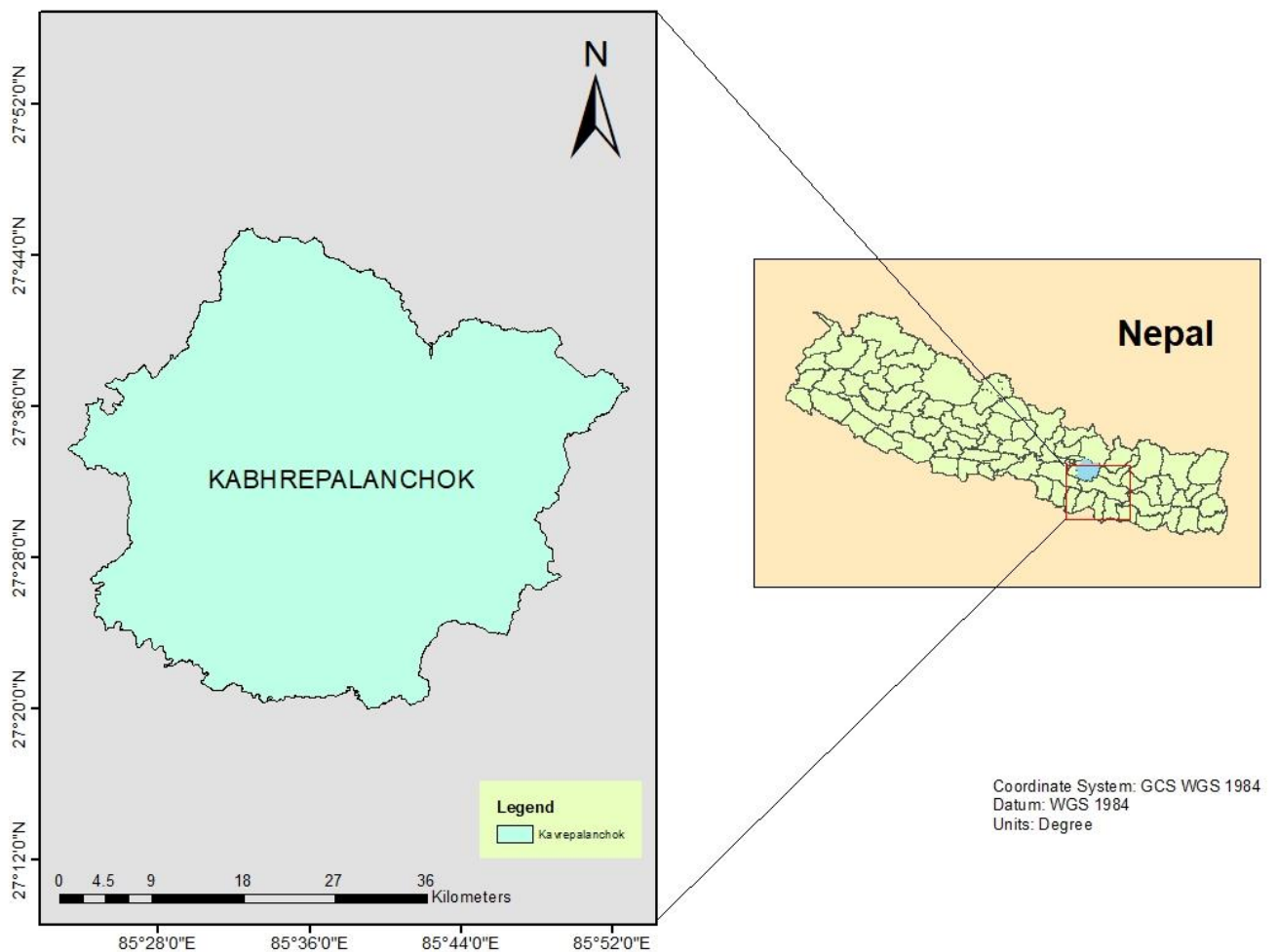


Figure 1: Study Area Map of Kavrepalanchok District

3. METHODOLOGY

Under this suitability analysis, we gathered the nine factors that influence in suitability of land for agriculture. Those factors include slope, elevation, aspect, land use and land cover, temperature, precipitation, soil moisture, distance from road, and distance from the river.

A) Data Collection

-The DEM data of the 30m resolution of the relevant district was downloaded from <https://dwtkns.com/srtm30m/>.

-Slope and aspect were derived from the DEM data.

-The land use and land cover map of the district was extracted from <https://livingatlas.arcgis.com/>.

-The data on distance from the road and distance from the river was downloaded through <https://extract.bbbike.org/>.

-The soil moisture of Kavrepalanchok District was clipped from the data available on <https://www.earthdata.nasa.gov/>.

-The average temperature and precipitation data of 30 seconds was extracted from <https://www.worldclim.org/>.

B) Weightage calculation and Score allocation

The influence percentage and suitability score of all the influencing factors were provided based on self-understanding and logical analysis.

Table 1: Table showing the weightage of the nine criteria.

S.N	Main Criteria	Weightage	Influence (%)	Sub-Criteria	Score/Rank
1.	Slope	0.15	15	0 to 3	10
				3 to 10	8
				10 to 20	6
				20 to 30	4
				30 to 89	2
2.	Elevation	0.05	5	954-1349	10
				1350-1744	8
				1745-2139	6
				2140-2534	4
				2535-2931	1

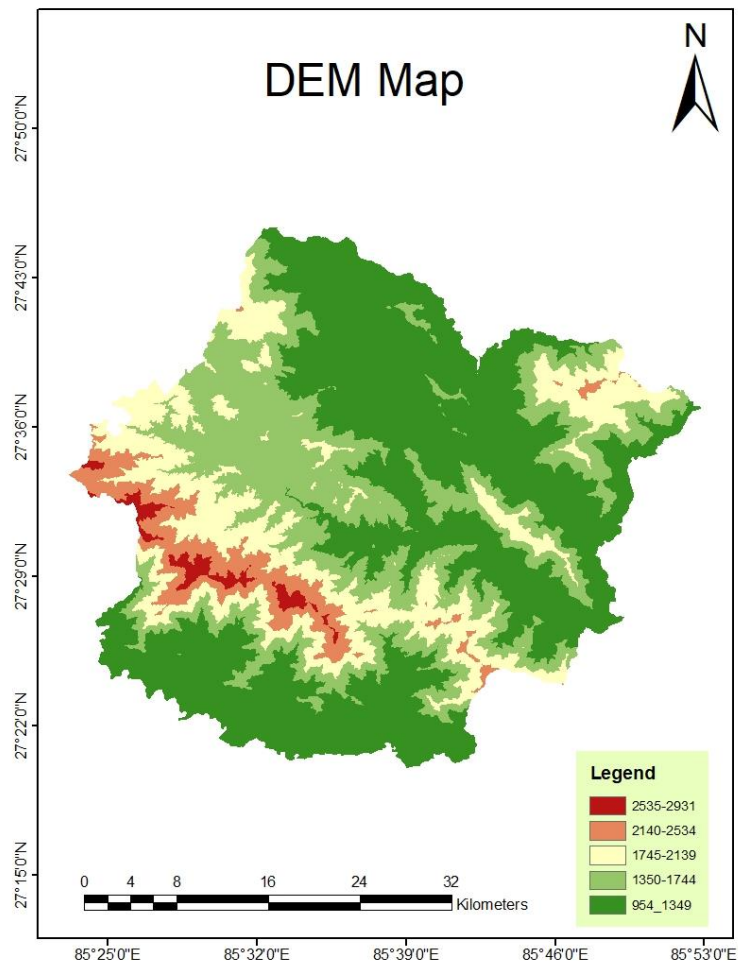
3.	Aspect	0.05	5	North-Northeast	5
				East-Southeast	10
				Southeast-South	8
				South-Southwest	7
				West-Northwest	3
				Northwest-North	2
4.	LULC	0.19	19	Water	1
				Trees	1
				Crops	10
				Built area	1
				Bare ground	5
				Range land	7
5.	Distance from RIver	0.08	8	0 to 435	10
				435 to 953	8
				954 to 1604	6
				1605-2639	4
				2630 to 5836	2
6.	Distance from Road	0.12	12	0 to 148	10
				149 to 381	8
				382 to 792	6
				793 to 1471	4
				1472 to 2971	2
7.	Soli Moisture	0.12	12	0 to 0.071	2
				0.07 to 0.14	4
				0.14 to 0.21	6
				0.21 to 0.33	8
				0.33 to 0.544	10
8.	Precipitation	0.12	12	67 to 70	2
				70 to 72	4
				72 to 74	6
				74 to 77	8
				77 to 82	10
9.	Avg Temperature	0.12	12	0 to 15	2
				15 to 17	4
				17 to 19	6
				19 t 21	8
				21 to 23.21	10

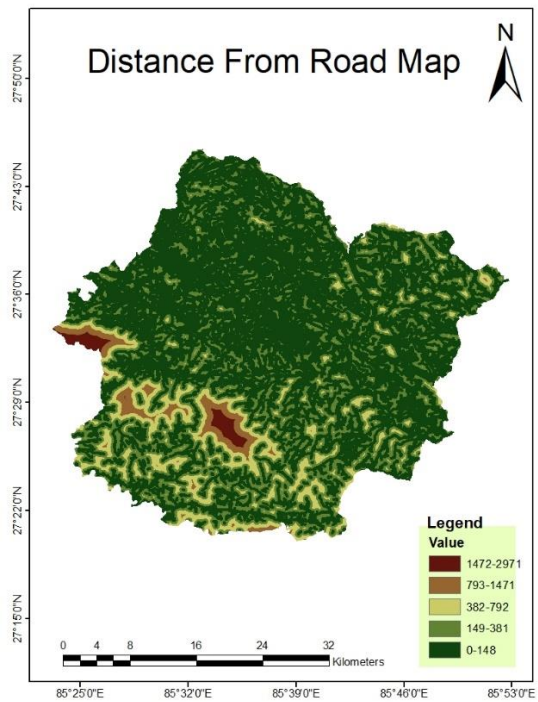
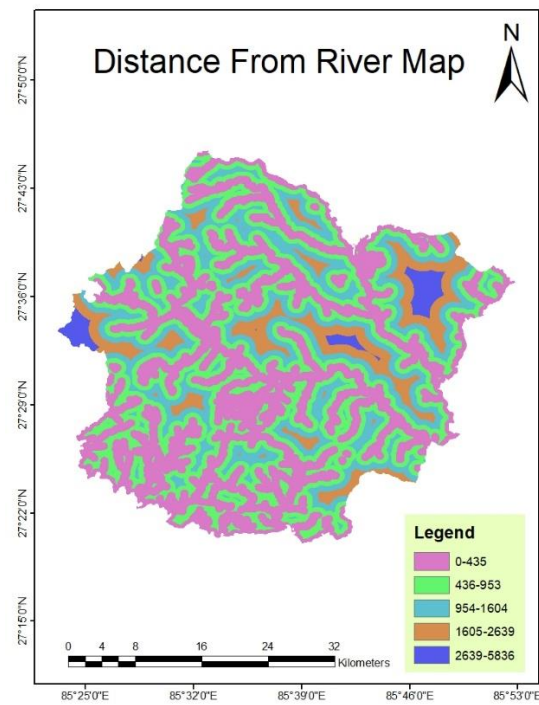
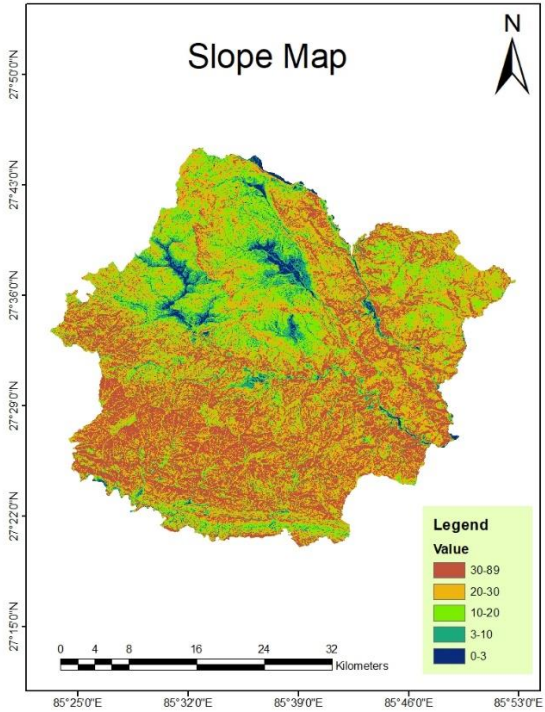
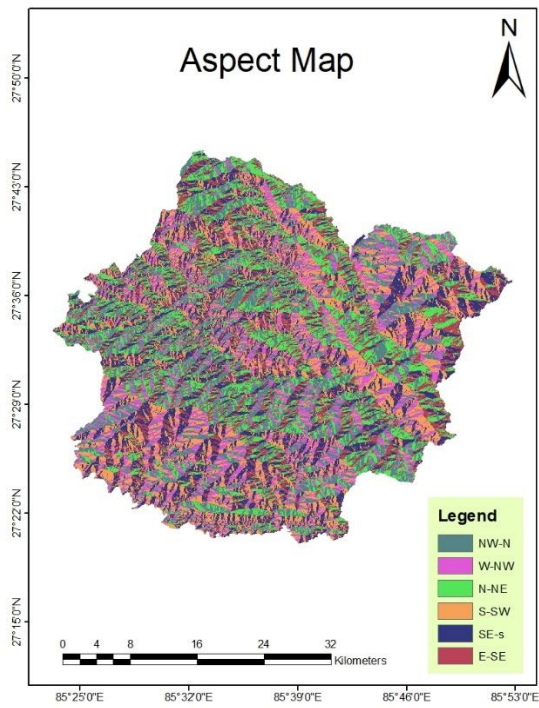
The score/rank given in Table 1 is based on Table 2 given below:

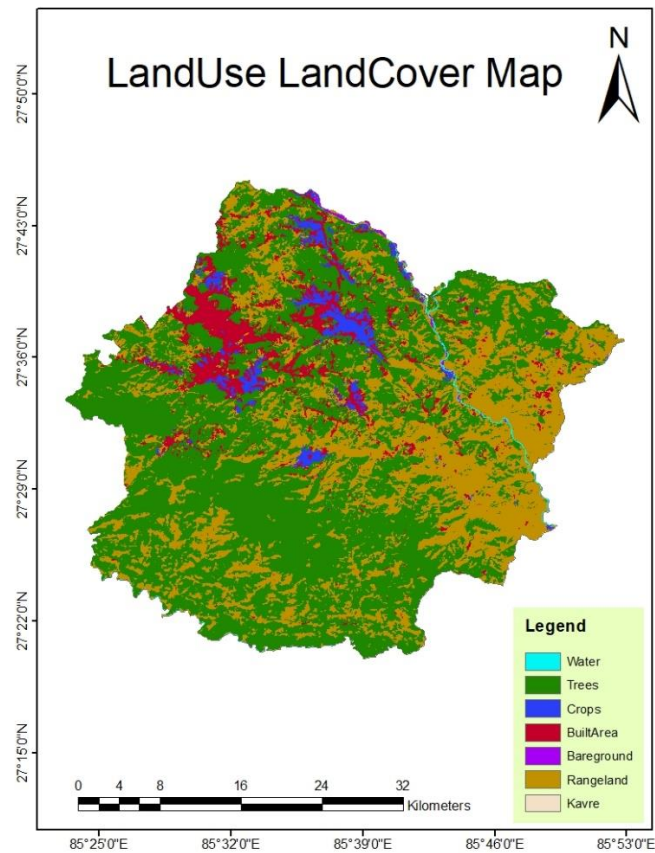
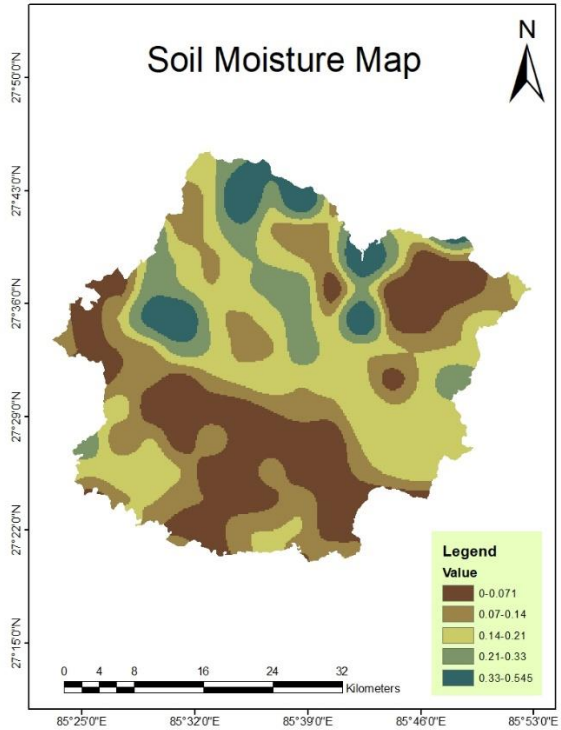
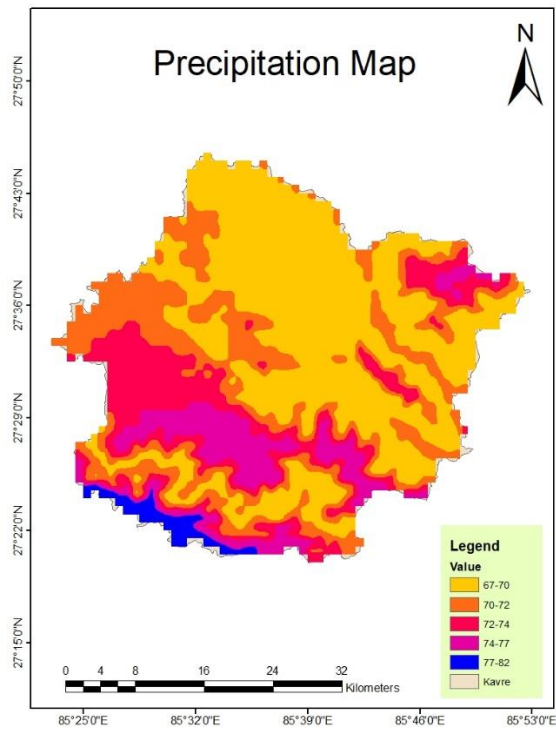
Table 2: Table showing score with their judgment

Verbal Judgment	Score/Rank
Extremely suitable	10
	9
Very strongly suitable	8
	7
Strongly suitable	6
	5
Moderately suitable	4
	3
Not suitable	2
	1

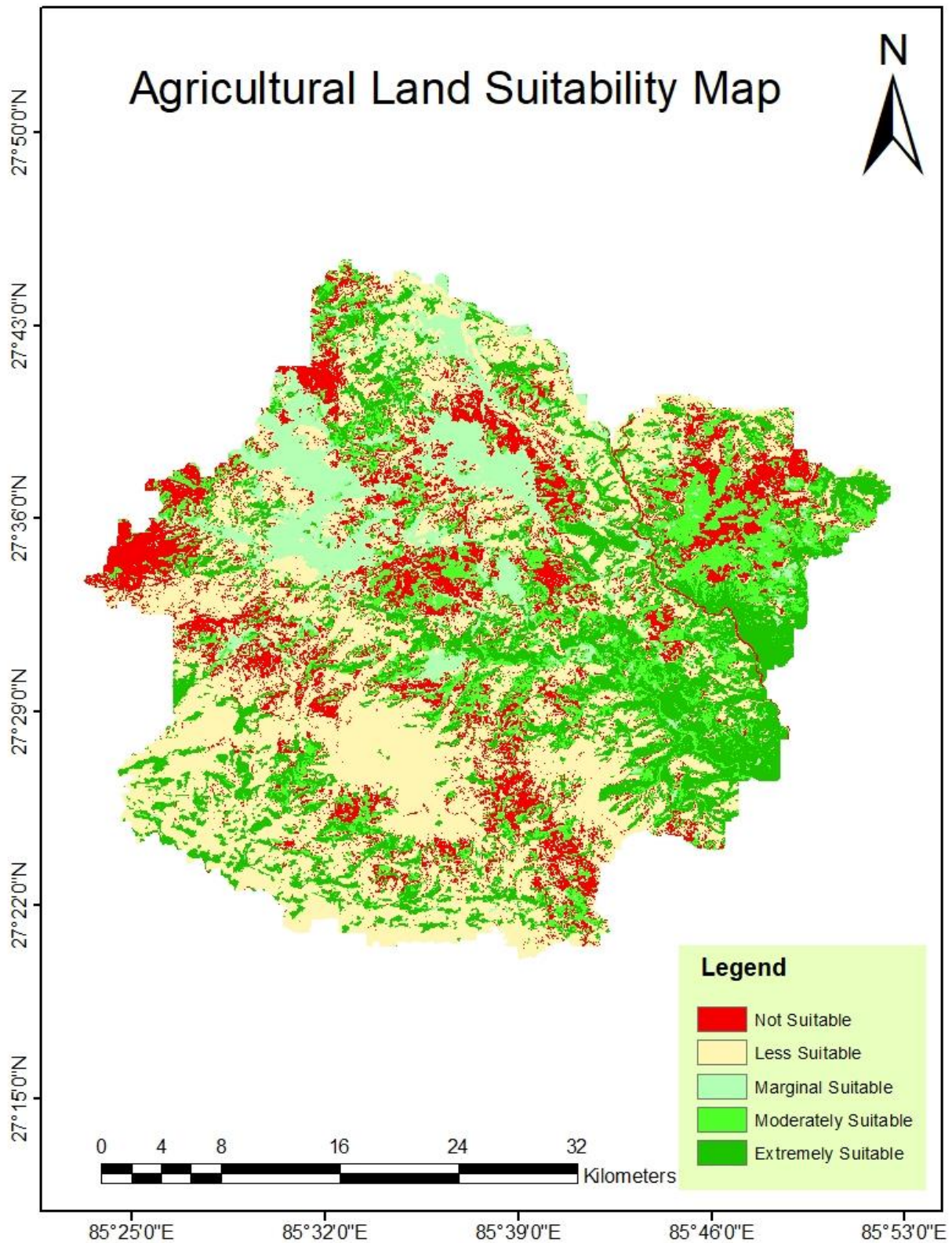
4. RESULTS







Final Map after Weighted Overlay Analysis



5. FINDINGS AND CONCLUSION

Table 3: Table showing area with percentage of suitable zones

S.No	Agricultural Land Suitable Zones	Area	
		Sq.km	Percent
1.	Extremely Suitable	291.914	21.04
2.	Moderately Suitable	164.369	11.85
4.	Marginal Suitable	145.544	10.49
4.	Less Suitable	776.944	56.01
5.	Not Suitable	8.330	0.60

In this study, we have taken the nine constituting factors namely slope, elevation, aspect, land use and land cover, temperature, precipitation, soil moisture, distance from road, and distance from the river that contribute to land for crop production. The weightage of those criteria was calculated based on logical analysis. After performing a weighted overlay, 21.04% of the land of Kavrepalanchok District is extremely suitable for agriculture while 11.85% is moderately suitable and 10.49% of the land is marginally suitable for crop production. The rest 56.61% of the area is less and not suitable for agriculture.